

ASSIGNMENT #2

Course: CHE 341. CHEMICAL PROCESS CONTROL
Term: Spring 2020
Professor: Dr. Michael Caracotsios
Office: EIB 244
E-mail: mcaracot@uic.edu

Your name: Adisa Hadzic

Due Date: Wednesday February 07 by 5:00pm in my office

Number of points.....100

Your score.....

ASSIGNMENT TOPIC

Review of Sensors Measuring Flow

Note: All problems are worth 100 points. Your final score will be the average value

Introduction

Flow control is essential in any chemical engineering process. Typical production begins with the flow control of the storage tank and its feed stream and continues to the control valve of the pump or the compressor, for example.[1] This type of control is used in obtaining of material balances, energy balances, process safety and process monitoring. [2] Feed stream flow control can be used to find and satisfy the material balances for a chemical process, since this balance is a first step in any process formation. In order to achieve the process that will be commercially successful, all listed categories need to be satisfied and the appropriate control and measurements applied. This type of control is continuously present in every system and it is regulated with flow sensors or flow meters.

Flow Sensors and Measurement

Flow measurements are critical in chemical processes, so that adequate selection of a flow transmitter that will translate sensors signals into measurements needs to follow subsequent guidelines: First, the phase of the flowing material as liquid, gas, solid or multiphase. Next, mass or volumetric flow measurements, as well as corrosiveness, accuracy, space available and cost.[2]

On-Line measurement objects and options used for flow process control are orifices, venturi, rotameter, turbine, vortex-shedding, ultrasonic, magnetic, thermal mass, Coriolis and target.[2] The measurement objects need to be sized for specific processes so they can provide ideal results.

Volumetric flow rate is measured by volumetric flow meter (positive displacement or PD flow meter), and it is computed by pressure drop across a length of a pipe or an orifice,

$$Q = \sqrt{\Delta P}$$

To measure pressure drop, standardized differential pressure (DP) instrumentation is used. For positive displacement flow meters, the signal sent to the output is proportional to the volume that is traveling through the meter. PD meters include bi-rotor types as gear, oval gear, helical gear, nutating disc, reciprocating piston and oscillating or rotary piston.[4]

Mass flow meters output signal is directly proportional to the mass traveling through the meter. They include thermal mass meters and Coriolis meters and they do not depend on pressure, temperature, viscosity and density. For low flow rates of the semiconductor manufacturing and the bioprocessing, mass flow controllers (MFC) are used to measure head loss.[2]

Velocity flow meters output signal is directly proportional to the velocity of the material traveling through the meter, or the change in kinetic energy. They can be electromagnetic, ultrasonic, turbine, propeller and paddle wheel, vortex shedding, target, variable area and rotameter, orifice plate, open channel, flow nozzle laminar, venturi and pitot tube objects.[4]

Importance of Flow in Industrial Applications

Flow control devices are very important in variety of industries because they measure the quantities of liquid or gas moving through the pipes with specific flow rates. There are hundreds of industries that use flow controls for volumetric fluid flow measurements, and they consist of chemical, refining, oil and gas, water/wastewater, electricity, pharmaceuticals, food and beverage and others.[5] Flow is important because low or high flow rates can breach safe limits of units in the process. There are multiple examples of plants disastrous accidents where flow valves failed to operate and compromised flow control. Flow meters are present in oil and gas industries, where large quantities of materials are transported between units. Coriolis meters are used in liquid and gas processes of nuclear installations, pharmaceutical industries, wastewater treatment plants and natural gas processing and transportation.[5] Differential pressure flow meters are used in chemical manufacturing, laboratories and flow control of gases using chromatography.[5] Gear flow meters are used is the flow control of liquids with high viscosity in hydraulics, transfer of fuels or oils, and production of paper.[5] Magnetic flow meters are used in technologies that uses static equipment and it is a state of art measuring device used in petrochemicals, chemicals, mining, pulp and water, water purification, food and beverage industry.[5]

Measurement Difficulties

Many different aspects can influence measurement system performance, as the correct instruments installation, calibration, optimal secondary instruments and fluid properties classification. [3] The additional factors that can cause difficulties in the measuring process are contamination or secondary phases inside the flow, solid particles presence, as well as the mechanical wear, corrosion and environmental influences. [3] Example of the such difficulties can be found in pharmaceutical industries where turbine, ultrasonic and Coriolis flow meters are used. The timing and exact accuracy must be present in production, since final products as medications have great impact on human health if any errors occurred during the process. The feature that is used to control the measurement is traceability, and it tracks a devices operation from calibration to use, to control necessary standards and track operative background of device. To avoid measurement difficulties, it is best to employ flow control systems based on their compatibilities and avoid low quality products.

Flow Control Loops

Flow control loops are fast systems used to respond to the disturbances rapidly, so that time between sampling and response is very short. Although flow control loops are at least difficult to use in comparison to the temperature, pressure and level loops, they are greatly included loops used for any continuous (or non-continuous) process where the flow of materials is essential.[6] This type of loops are not too complex to operate but they are the base of any process control system. For example, flow control loops are used as an inner loop of a cascade control system, and their behavior is usually non-linear.[6]

Industrial flow control loop is presented in Figure 1, and it consists of a flow transmitter (FT), flow controller (FC), flow sensor, control valve (FCV), and a pump. Control valve controls the flow, while pump delivers more flow when needed. The flow inside the pipe is measured with devices mentioned above as orifice, venturi or ultrasonic flow. The measured value is then transmitted using FT and sent to a controller FC, where value is compared to the setpoint value. The difference of the two is recorded and sent to the control device that corresponds to the results and changes flow if necessary. If the measured value is higher than a setpoint then the flow is decreased or blocked. However, if the measured value is smaller than the setpoint value the control valve opens to increase the flow. The figure below shows simple water flow control loop.

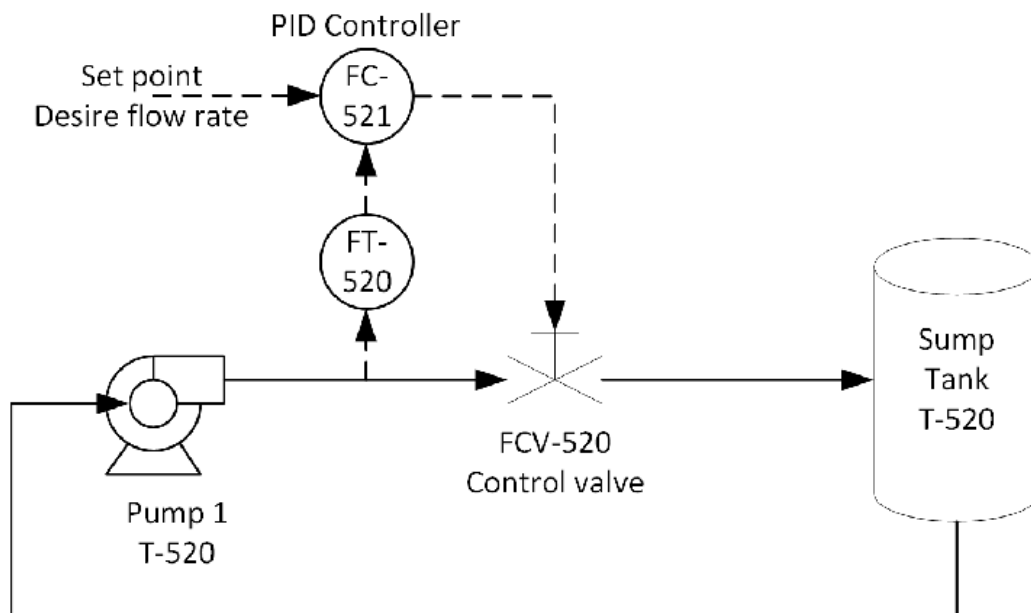


Figure 1: Industrial flow control loop [7]

References

- [1] Towler, Gavin, et al. *Chemical Engineering Design: Principles, Practice, and Economics of Plant and Process Design*. Elsevier, 2013.
- [2] Seborg, Dale E., et al. *Process Dynamics and Control*. Wiley, 2017.
- [3] www.flowcontrolnetwork.com/instrumentation/flow-measurement/article/15564230/the-importance-of-accurate-flow-measurement-in-industrial-processes.
- [4] “What Is a Flow Meter?” *Max Precision Flow Meters*, www.maxmachinery.com/what-is-a-flow-meter/.
- [5] Admin. “The Importance Of Flow Meters In Industries.” *Every Thing For Dads*, Every Thing For Dads, 6 May 2019, www.everythingfordads.com/dadsblog/the-importance-of-flow-meters-in-industries.
- [6] *Flow Control*, www.cybosoft.com/solutions/flowcontrol.html
- [7] https://www.researchgate.net/figure/SE113-Water-Flow-Control-Schematic-Diagram_fig1_332432639